

KANCHANAMALA P

✉ malakanchana26@gmail.com ☎ 7299668990 📍 CHENNAI [in Kanchanamala P](#) [🔄 kanchanamala-dev](#) [🔗 Portfolio Website](#)

PROFILE

Second-year Electronics and Communication Engineering student with strong interest in problem-solving, embedded systems, and software development. Motivated to learn industry-level engineering practices and contribute through hands-on projects combining hardware and software.

LEADERSHIP AND RESPONSIBILITIES

08/2024 – 08/2025
VIT CHENNAI **Programme Representative – Electronics and Communication Engineering (ECE)**
Represented ECE students in academic and departmental matters, strengthening leadership, responsibility, and teamwork through addressing concerns and collaborating with faculty and peers.

EDUCATION

2024 – 2028
CHENNAI **B.Tech in Electronics and Communication Engineering, VELLORE INSTITUTE OF TECHNOLOGY**
CGPA: 9.32 / 10

2023 – 2024
AVADI **Senior Secondary, JGVV (CBSE BOARD)**
Score: 458 / 500 (91.6%)

2021 – 2022
AVADI **Secondary, JGVV (CBSE BOARD)**
Score: 473 / 500 (94.6%)

SOFT SKILL

- Time Management
- Problem Solving
- Attention To Detail
- Communication Skills

CERTIFICATIONS & WORKSHOPS

IoT Automation with Raspberry Pi – IIT Madras [🔗](#)
• Participated in a hands-on workshop covering **NodeMCU, Raspberry Pi basics, and IoT automation concepts.**

Unlocking the IoT Universe: Hands-on Masterclass – VIT Chennai [🔗](#)
• Gained practical exposure to **NodeMCU interfacing and core IoT workflows** through guided sessions.

PROJECTS

12/2025 – 04/2026 **Whistle-Based Smart Gas Stove Automation, Embedded Systems Project**
• Developed an STM32-based system to detect cooker whistles and count user-defined targets.
• Automated gas knob control using motor actuation for kitchen safety.

05/2025 – 04/2026 **Smart Home Energy Monitoring & Prediction Dashboard, IoT & Machine Learning Project**
• Built a real-time energy monitoring system using Raspberry Pi, NodeMCU, InfluxDB, and Grafana.
• Implemented a Python ML model to forecast household energy consumption.

07/2025 – 12/2025 **Morse Code Translator Using FPGA, FPGA & Digital Design Project**
• Designed a Verilog-based FPGA system to convert Morse button inputs into ASCII characters.
• Implemented timing logic for accurate symbol differentiation and decoding.

07/2025 – 12/2025 **Cell Phone Detector Using Analog Components** [🔗](#) , *Analog Circuits & PCB Project*
• Built a mobile signal detection circuit using antenna and LM357 op-amp.
• Designed PCB and performed component placement, soldering, and debugging.

07/2024 – 12/2024 **IoT-Based Plant Monitoring and Watering System** [🔗](#) , *Embedded IoT Automation Project*
• Developed automated irrigation using NodeMCU and environmental sensors.
• Enabled remote monitoring via mobile dashboard.

07/2024 – 12/2024 **Currency Detector and Total Amount Calculator** [🔗](#) , *Embedded Systems & Sensors Project*
• Designed IR/color sensor-based Indian currency detection system.
• Implemented logic for note classification and total amount calculation.

12/2025 – 04/2026 **Junctionless Double Gate MOSFET Design & CMOS Analysis, TCAD & VLSI Simulation Project**
• Simulating JLDG NMOS/PMOS in Synopsys Sentaurus and analyzing device characteristics.
• Preparing matched devices for CMOS inverter implementation in Cadence Virtuoso.

INTERESTS

Internet of Things (IoT) | Embedded Systems | Automation and Control Systems | Analog and Mixed Signal Circuits

EXPERIENCE

05/2026 – 07/2026
Chennai **R&D Intern, Jedlik Motor Pvt. Ltd.**
• Worked on STM32-based embedded motor control systems.
• Performed motor profiling, firmware deployment, and system validation for BLDC/PMSM drives.
• Worked on FOC based design of BLDC/PMSM motor drivers and control electronics.

TECHNICAL SKILLS

JAVA | Python | Embedded C | Verilog HDL | PCB Design (EasyEDA) | Assembly Programming | GRAFANA | MATLAB

LANGUAGES

ENGLISH - Proficient | TAMIL - Native | FRENCH - Basic